



## Tangent ratio

- 1 Match the estimated values with the trigonometric functions below. Write the correct letter in each box

|   |                  |
|---|------------------|
| a | $\sin(23^\circ)$ |
| b | $\cos(39^\circ)$ |
| c | $\tan(22^\circ)$ |
| d | $\tan(45^\circ)$ |

|   |       |
|---|-------|
| a | 0.39  |
| b | 0.777 |
| d | 1     |
| c | 0.404 |

- 2 For which value of  $\theta$  is  $\tan(\theta)$  undefined? Fill in the blank

90

- 3 In a right triangle, if you are provided with  $\theta$  and the opposite side, which function would you use to find the adjacent side? Tick **1** correct answer

- ☐ Sine
- ☐ Cosine
- ☒ Tangent

- 4 Which of these is a rearrangement of  $\tan(\theta) = \frac{\text{opp}}{\text{adj}}$  Tick **1** correct answer

- ☐  $\tan(\theta) = \frac{\text{adj}}{\text{opp}}$
- ☐  $\text{opp} = \frac{\tan(\theta)}{\text{adj}}$
- ☒  $\text{adj} = \frac{\text{opp}}{\tan(\theta)}$



5 Which of these is not an rearrangement of  $\tan(\theta) = \frac{\text{opp}}{\text{adj}}$  Tick **1** correct answer

☐  $\text{adj} = \frac{\text{opp}}{\tan(\theta)}$

☐  $\tan(\theta) \times \text{adj} = \text{opp}$

☒  $\tan(\theta) = \frac{\text{adj}}{\text{opp}}$

6 For a right triangle ABC, if angle CBA =  $33^\circ$ , and the adjacent side to that angle is 1 cm, what is the length of the side opposite angle CBA? Tick **1** correct answer

☐ 0.545 cm (3 d.p)

☒ 0.649 cm (3 d.p)

☐ 1 cm

